

Element E

Our proposed solution integrates several high-priority STEM principles and practices to meet the design requirements effectively. We utilized engineering principles to stabilize the frame and enhance animal comfort by incorporating an engine leveler. This provides a fulcrum to balance the weight towards the animal's chest, reducing pressure on the gastrointestinal system (Rodriguez, 2024). Furthermore, to improve weight distribution, we added multiple points of contact with the tarp provided by the zoo, reducing stress on the system and offering a more level support system for the animal.

We selected Superstrut, a galvanized steel material, for its superior strength and resistance to corrosion compared to the previously used aluminum bars. The zinc coating on the steel ensures durability and firmness, essential for supporting heavy animals. The width of the upper frame was also optimized to affect weight distribution positively, enhancing the security of the animal during transportation. Additionally, we identified a single point of contact to ensure a stable and consistent connection, replacing the less reliable wire clamp on zipline cables.

The proposed solution is well substantiated with STEM principles and practices. The engineering design addresses the stabilization of the frame and animal comfort using an engine leveler, which applies principles of mechanical advantage and load distribution (Rodriguez, 2024). The use of Superstrut material incorporates materials science, ensuring the structure is robust and corrosion-resistant. The consideration of the frame's width and the single point of contact demonstrates an application of structural engineering principles to maintain balance and stability, ensuring the animal's safety.

Our design has been thoroughly reviewed by Bill Rodriguez, a field expert from Vanderbilt. Mr. Rodriguez has spent years studying and practicing many branches of engineering and has dedicated time to mechanical engineering projects that require similar criteria as does our project. His feedback highlighted the importance of using an engine leveler for load balancing and the advantages of Superstrut over aluminum bars in terms of strength and corrosion resistance. His insights on optimizing the frame's width for better weight distribution and secure connection points were instrumental in refining our design. These reviews confirm that the design aligns with relevant STEM principles and practices.

Additionally, our current design was reviewed by a graduate student from Vanderbilt named Lucas, as well as award-winning engineer Jeremy Qualls. Both of these field experts have devoted their time and energy to studying and furthering their knowledge--and the knowledge of others around them--on the engineering field. Jeremy Qualls made a few suggestions regarding the safety of the animals in an emergency situation and the accessibility of the fulcrum-adjuster of the engine leveler. Specifically, he suggested that we integrate an “emergency release system” into our prototype in the event of an animal becoming stressed and panicking while in the lift. However, upon conferring with our primary stakeholder, Dr. Wancea, we have decided that the animals utilizing the lift will either be under anesthesia or ill to the point of extreme lethargy, reducing the risks of any circumstances similar to what was described by Mr. Qualls.

The design review conducted with Bill Rodriguez resulted in clear and detailed confirmation of the proposed solution. He verified the effectiveness of using an engine leveler and Superstrut material and suggested improvements for weight distribution and stability. His feedback has been instrumental in making necessary corrections and optimizations, ensuring the design is robust and reliable.

Citations:

- Rodriguez, B. (2024). Personal interview. "Insights on Animal Lifting Mechanism Design." Vanderbilt University.
- Wancea, S. (2024). Personal Interview and Design Review